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Pre-Miocene palaeogeography of the Los Cabos Block, Baja California Sur: geochronological and palaeomagnetic constraints

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Abstract

The Los Cabos Block (LCB) is located at the southern end of the Baja California peninsula and is composed mainly of intrusive rocks that were emplaced into pre-Cretaceous heterogeneous metasediments and are partly covered by Miocene volcanic and volcaniclastic rocks. Field observations provide evidence that the undeformed, homogeneous gabbronorite of the Sierra El Novillo in the NE part of the LCB is the oldest crystalline unit, which is intruded by undeformed to slightly deformed tonalitic-quartz-dioritic granitoids. For these rocks an Rb–Sr intrusion age of 129 ± 15 Ma and a biotite whole-rock cooling age of 116 ± 2 Ma were obtained. Low initial $^{87}Sr/^{86}Sr$ and high ϵ-Nd values of about 0.7036 and $+5$ to $+6$ respectively characterise the intrusives as primitive, mantle-derived granites. Further to the southeast and south, granitoids are more silicic, with abundant deformed sequences — orthogneisses, diatexites and migmatites, among others. From the southern part of the LCB an Rb–Sr intrusion age for the undeformed granites of 115 ± 4 Ma and a biotite cooling age of 90 ± 2 Ma were determined. Initial $^{87}Sr/^{86}Sr$ was determined to be 0.7054 and ϵ-Nd values fall between -2 and 0. The isotopic data do not clearly allow us to characterise the deformed granitoids of the LCB as being intruded by the undeformed unit. Deformation could have been partly due to syn-intrusive tectonics.

Palaeomagnetic data suggest minor, if any, northward displacement of the LCB with respect to continental Mexico, corresponding to the rifting in the Gulf of California since the late Miocene, and significant 35–45° clockwise rotations. Possible tilting effects, which may have occurred given the dip of Tertiary volcanics on top of the intrusives, would modify these values for rotation and to a lesser degree of northward displacement.

Geochemical, isotopic and palaeomagnetic results and the cooling history of the LCB are similar to those from the Puerto Vallarta Batholith, whereas other Mexican continental margin plutons further to the SE display lower intrusion ages, more rapid cooling rates and generally more primitive magma compositions. In the light of the data presented, a common magmatic evolution of the LCB and the Puerto Vallarta Batholith is strongly suggested. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: Baja California; geochronology; palaeogeography; palaeomagnetism

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1. Introduction

Palaeogeographic reconstructions for the Baja California peninsula have yielded very controversial models in papers published over the past 20 years [see reviews in Böhnel et al. (1992) and Dickinson and Butler (1998)]. Most hypotheses place the peninsula at lower palaeolatitudes — in front of Central America or of southern Mexico — with subsequent 10 to 20° northward translation(s) between late Cretaceous and Miocene. Many models include clockwise rotations from 25 to 45°.

It is also not clear whether Baja acted as a uniform unit during that translation, or if it is composed of lithologically distinguishable blocks with different ages and origins. If the latter scenario is more likely, what is the provenance and time of accretion of the crustal mosaic that constitutes the present day peninsula of Baja California? Based on the available geophysical and geological evidence, Sedlock et al. (1993) proposed a detailed palaeogeographic model for Baja California, involving large-scale northward and southward displacements during the Mesozoic.

The main reason for the hypothetical character and contradictory nature of palaeogeographic models is the continuing lack of geological fieldwork and mapping for many areas of Baja California. On the other hand, there is not enough interdisciplinary collaboration among structural geologists, geochronologists, palaeontologists and palaeomagnetists. Many working groups are concentrating their investigations within their discipline and use, for instance, age data just as numbers to support their theory without a critical evaluation of the quality of these data.

In this paper we present Rb–Sr and Sm–Nd isotopic data and the results of palaeomagnetic investigations on intrusive rocks from the southern tip of Baja California, between La Paz and Cabo San Lucas, referred to herein as the Los Cabos Block (LCB). The aim of our work was also to compare these results with previous data from the Mexican continental magmatic arc.

2. Geology and petrography

The most abundant rocks of the LCB are undeformed intrusives with granitic to tonalitic compositions (Fig. 1), which are found mainly in the south-central part of the peninsula, including the area around Cabo San Lucas. Plagioclase (oligoclase) + quartz + alkali feldspar + biotite + horn blende are the principal rock-forming minerals, with white mica, apatite, sphene, zircon, magnetite and other opaque minerals as common accessories.

Another rock unit is represented by the gabbronorites of the El Novillo Complex, about 30 km south of La Paz. It is a quite homogeneous, nearly undeformed, lens-shaped body of $\sim 75\ km^2$ with plagioclase (andesine), ortho- and clino-pyroxene as major minerals and accessory biotite, hornblende, quartz and opaque minerals (magnetite and ilmenite). Biotite and quartz are only conspicuous at the rims of the complex, which is intruded partly by the undeformed LCB granitoids and partly by a third important unit: moderately to strongly deformed granitoids.

Deformed granitoids are generally more abundant in the central and northern LCB, but they also occur in the central and eastern part of the southern LCB. Their composition is quite heterogeneous, ranging from granites, granodiorites, tonalites, hornblende-diorites and hornblende-gabbros as the main constituents, to some minor cumulative pyroxenites and hornblendites south of the El Novillo Complex (Riewe, 1998). Xenoliths of grabbronoritic composition, which are interpreted to come from the El Novillo unit, were found in the deformed granitoids, thus giving some field evidence for the Novillo Complex to be the oldest intrusive unit of the LCB. On the other hand, field observations do not clearly permit an age assignment of the deformed plutonic rocks with respect to the undeformed ones.

Another interesting observation is the presence of volcanosedimentary inclusions in the deformed tonalites and quartz-diorites southeast of the Sierra El Novillo (Riewe, 1998).

Both deformed and undeformed granitoids are cut by younger, basaltic to andesitic dikes of unknown age. Miocene volcanic and volcaniclastic rocks overlie the crystalline basement in parts of the eastern LCB, for instance on the Espíritu Santo islands (Aranda-Gómez and Pérez-Venzor, 1986). The roof-pendant of the plutonic rocks is com-

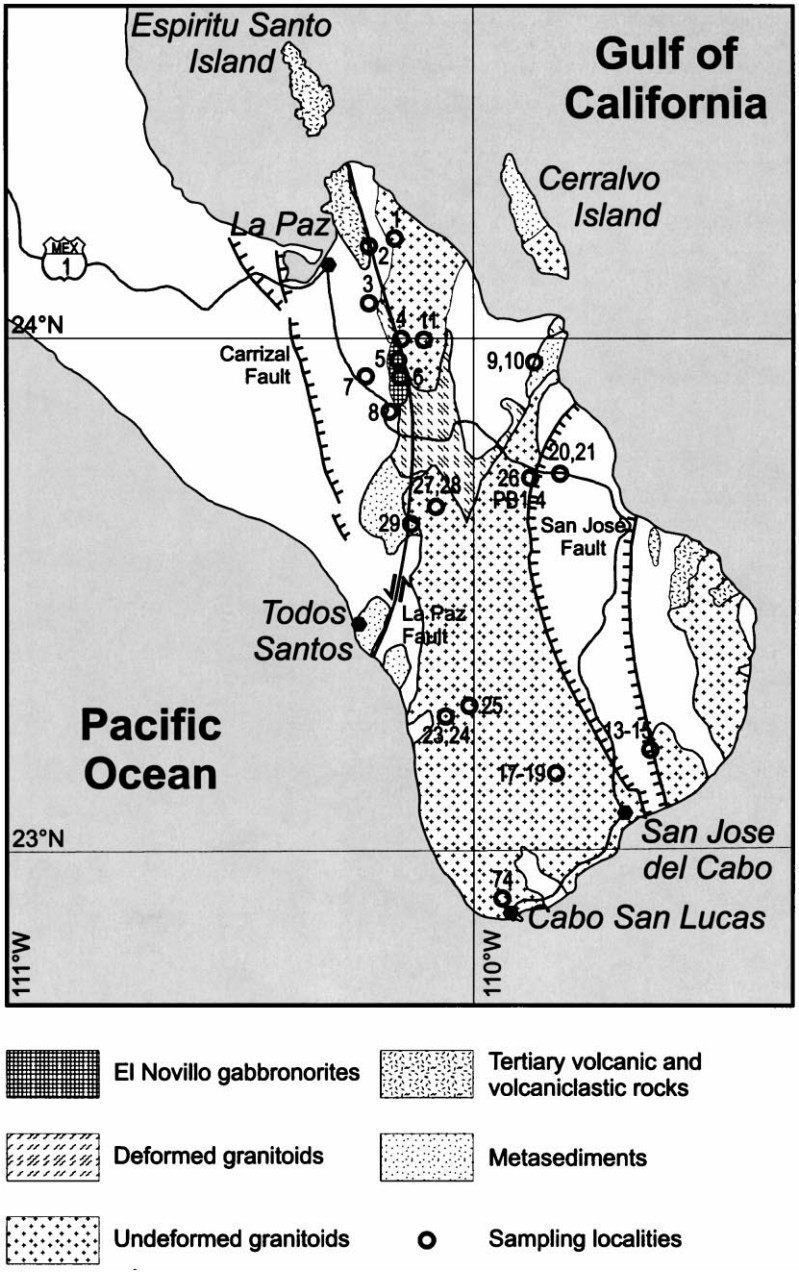


Fig. 1. Simplified geological map of the LCB, Baja California Sur, with the lithological distribution of the different magmatic rock units, the metasediments and sampling locations and numbers. Alluvium and Tertiary sediments are shown in white.

posed of metasediments with skarns, marbles, slates, mica- and garnet-bearing schists, paragneisses and orthogneisses, among others.

The main tectonic elements are the nearly N–S-striking La Paz Fault, a normal fault system that dips to the west and has been repeatedly reactivated, and the San José Fault, both of unknown but probably Cenozoic age. Younger,

generally E–W-trending strike-slip faults and Horst and Graben structures in the central part of the LCB are related to the opening of the Gulf of California. More detailed geological descriptions of the north-central part of the LCB are given by Aranda-Gómez and Pérez-Venzor (1988, 1989).

According to Campa and Coney (1983), the LCB is part of the Alisitos terrane, which is mostly composed of plutonic rocks of the Peninsular Ranges Batholith (PRB) in central and northern Baja. In a more recent paper by Sedlock et al. (1993), the LCB forms an independent tectonostratigraphic unit, named Pericú terrane. In some Cretaceous palaeogeographic reconstructions, the LCB is situated NW of the Puerto Vallarta Batholith (Gastil, 1983; Sedlock et al., 1993), which also results in a good fit of the actual coastlines. Nevertheless, this hypothesis is lacking supporting data.

3. Previous isotopic and palaeomagnetic work

There is not much material to be found in the literature concerning age determinations of the intrusive rocks of the LCB, and what there is is mainly published as abstracts and without analytical details. Most age data have been obtained by the K–Ar method. Ortega-Gutiérrez (1982) presented a K–Ar mineral age for granitoids near Todos Santos of 74 Ma. Hausback (1984) reports a 115 ± 2.4 Ma K–Ar hornblende age from a deformed hornblende-diorite. Frizzell et al. (1984) obtained from an undeformed granite in the southern LCB a K–Ar mineral age of 93 Ma and a U–Pb zircon age of 94 Ma. Other unpublished 115–65 Ma K–Ar mineral age determinations by Frizzell et al. (1984) are cited in Aranda-Gómez and Pérez-Venzor (1989). Murillo-Muñetón (1991) presented K–Ar biotite, muscovite, and hornblende ages between 91 and 64 Ma from undeformed samples and a 116 ± 6 Ma hornblende age from the deformed unit.

There are some indications of the existence of Jurassic–Permian intrusive activity in the LCB (Altamirano, 1971; Gastil et al., 1978), again without analytical data and descriptions of the sample localities.

Although many palaeomagnetic data have been published for Mesozoic rocks from Baja [see review in Hagstrum and Filmer (1990)], only four sites have been sampled from the LCB (Hagstrum et al., 1985). These come from the vicinity of San Bartolo, $\sim$40 km SE of La Paz. One of the sites did not yield reliable data, and the others are included in Table 3, together with our new data. Hagstrum et al. (1985) calculated a palaeopole for the PRB, using the three sites from the LCB and another four sites from the Cataviña batholith in central Baja. This palaeopole apparently was displaced 12° to the north and rotated 29° clockwise. Including 32 sites from the PRB in southern California, Hagstrum and Filmer (1990) published a palaeopole situated at 83.2°N, 189.8°E, which apparently was calculated from the mean of the 39 site-mean directions. This is not correct, as the sites extend over $\sim 10^\circ$ of latitude, and the palaeopole has to be calculated from virtual geomagnetic poles (VGPs). Doing so we obtained a palaeopole at 84.6°N, 346.1°E, $A_{95} = 3.2^\circ$, which diverges by 11.9° from the palaeopole given by Hagstrum and Filmer (1990). Comparing this palaeopole with the 100 Ma reference pole by Globerman and Irving (1988), the northward displacement is slightly reduced to $9 \pm 5^\circ$, with a clockwise rotation of $22 \pm 6^\circ$.

4. Sampling and analytical procedures

A total of 18 whole-rock (WR) samples were taken for isotopic investigations; 12 of these and another 21 sites were sampled for palaeomagnetic analyses, using a portable drill with diamond bits. Sampling localities are shown in Fig. 1, and site coordinates are given in Tables 1 and 3.

In general we avoided samples with chloritised biotite or other indications of chemical alteration that might suggest a perturbation of the Rb–Sr isotopic system. In the southeastern part of the LCB, east of the San José Fault system, and around Cabo San Lucas, the sampling of fresh and non-seawater altered material was difficult.

Since the Rb–Sr and Sm–Nd isotopic data presented in this paper are among the first results obtained in the new isotope laboratory (LUGIS)

Table 1
Major- and trace-element abundances of the LCB intrusives. Major elements (wt%), oxides, trace elements (ppm). –: no data; *: all Fe as Fe_2O_3; LOI: loss on ignition

	E La Paz	Mafic intrusives near El Novillo					Undeformed granitoids								Deformed granitoids			
	LCB 3 Tonalite Rancho Quemado 24°04′08″ 110°11′45″	LCB-5 Qz-diorite N Las Calabazas 23°57′38″ 110°09′30″	LCB-6 Qz-diorite Misión La Trinchera 23°53′01″ 110°09′09″	LCB-7 Tonalite Rancho El Novillo 23°55′40″ 110°14′30″	LCB-8A Gab.norite Presa Cerro 23°51′42″ 110°10′20″	LCB-8B Hbldite. Presa Cerro 23°51′47″ 110°10′16″	LCB-13 Bi-Granite Road Palo Escopeta 23°08′50″ 109°41′09″	LCB-18 Tonalite Presa San Lazaro 23°07′54″ 109°49′02″	LCB-19 A Bi-Granite Presa San Lazaro 23°07′54″ 109°49′00″	LCB-19 C Granodior. Presa San Lazaro 23°07′54″ 109°49′00″	LCB-20 Granodior. E San Bartolo 23°44′30″ 109°49′29″	LCB-21 Bi-Granite E San Bartolo 23°44′31″ 110°49′33″	LCB-23 Granodior. El Aguaje 23°44′30″ 110°03′00″	MS 74 Granodior. Cabo San Lucas 22°54′25″ 109°55′45″	LCB-1 Def. Granite Cajoncito 24°08′58″ 110°12′55″	LCB-9 Diatex. Granite S. La Gata 23°57′35″ 109°52′03″	LCB-17 Def. Gd. Presa San Lazaro 23°08′08″ 109°48′57″	LCB-19 B migm. Gd. Presa San Lazaro 23°07′54″ 109°49′00″
SiO_2	56.21	55.91	51.54	56.91	43.17	41.05	69.93	52.29	75.16	63.11	63.39	73.33	65.08	68.84	71.68	74.80	65.17	71.96
TiO_2	0.91	0.93	0.95	0.96	0.56	1.80	0.35	1.28	0.16	0.86	0.69	0.22	0.56	0.37	0.13	0.03	0.43	0.26
Al_2O_3	18.29	17.15	19.04	17.36	21.26	17.13	15.58	18.95	13.66	17.2	17.03	14.31	16.1	15.97	15.42	13.67	17.78	14.11
$Fe_2O_3^*$	8.67	9.62	10.26	8.72	10.54	14.41	2.89	8.95	2.17	5.81	5.16	2.71	4.68	2.45	1.91	2.08	3.38	2.70
MnO	0.13	0.16	0.16	0.13	0.10	0.15	0.03	0.13	0.03	0.07	0.05	0.04	0.06	0.04	0.02	0.09	0.04	0.04
MgO	3.32	3.94	5.06	3.41	8.32	9.57	0.76	4.26	0.18	2.34	1.93	0.35	2.04	0.80	0.43	0.11	0.93	0.54
CaO	6.47	7.94	9.09	6.92	14.67	11.44	2.38	8.06	1.36	5.10	4.59	1.99	4.30	2.40	2.02	0.95	3.76	1.20
Na_2O	3.99	3.26	3.45	4.37	1.39	1.88	3.93	4.02	3.02	3.12	3.71	4.49	3.61	4.19	5.12	3.48	5.16	2.94
K_2O	1.40	1.04	0.28	1.45	0.47	0.27	3.11	1.24	4.72	2.41	2.17	3.22	2.42	2.87	2.36	4.30	2.92	5.11
P_2O_5	0.17	0.11	0.14	0.15	0.00	0.14	0.12	0.23	0.00	0.16	0.20	0.02	0.16	0.13	0.06	0.04	0.07	0.14
LOI	0.10	0.03	0.02	0.03	−0.12	1.9	0.70	0.38	0.20	0.20	0.70	0.28	0.65	0.03	0.62	0.16	0.21	0.46
Total	99.66	100.09	99.99	100.41	100.36	99.74	99.68	99.79	100.66	100.38	99.62	100.96	99.66	98.09	99.77	99.70	100.42	99.45
Ba	381	166	248	473	31.2	91.0	877	785	1886	736	948	–	785	–	1052	400	1057	656
Rb	55	15.7	1.2	26	0.8	2.3	98	24	119	92.2	78	103	74	77	41	154	86	133
Sr	451	364	523	400	271	232	419	703	185	577	473	340	442	621	750	121	471	91
Y	10.55	12.48	10.72	18.69	3.84	16.9	11.5	15.15	5.66	10.6	14.7	–	9.74	–	3.06	15.5	8.24	16.7
Zr	–	–	–	–	–	24.6	145	–	–	–	157	–	120	–	61.6	43.4	–	116
Nb	3.37	2.39	1.61	6.29	0.41	1.20	5.73	5.75	6.00	6.54	6.61	–	5.62	–	2.04	7.76	6.15	4.91
Th	–	–	–	–	–	0.04	6.12	–	–	–	5.5	–	5.92	–	1.02	14.03	–	10.15
Pb	–	–	–	–	–	1.99	18.4	–	–	–	14.3	–	14.9	–	14.4	24.8	–	32.1
Ga	–	–	–	–	–	18.5	20.7	–	–	–	24.1	–	18.5	–	16.6	13.0	–	15.0
Zn	–	–	–	–	–	87.3	60.0	–	–	–	85.6	–	60.9	–	21.4	21.8	–	44.6
Cu	–	–	–	–	–	582	10.0	–	–	–	15.9	–	9.47	–	10.4	18.9	–	14.9
Ni	–	–	–	–	–	31.0	7.55	–	–	–	10.9	–	13.2	–	7.85	9.8	–	6.82
V	–	–	–	–	–	637	30.3	–	–	–	66.1	–	79.2	–	15.0	3.2	–	16.7
Cr	–	–	–	–	–	33.5	90.4	–	–	–	96.6	–	105	–	57.2	109	–	43.9
Hf	–	–	–	–	–	0.95	3.91	–	–	–	3.99	–	3.36	–	1.7	1.99	–	3.22
Cs	–	–	–	–	–	0.28	3.04	–	–	–	2.15	–	1.93	–	1.5	2.1	–	3.64
Sc	–	–	–	–	–	40.2	4.50	–	–	–	8.60	–	8.3	–	3.2	2.4	–	4.90
Ta	–	–	–	–	–	0.093	0.56	–	–	–	0.59	–	0.63	–	0.16	0.96	–	0.47
Co	–	–	–	–	–	57.0	3.31	–	–	–	8.81	–	9.44	–	2.18	1.52	–	2.40
Be	–	–	–	–	–	traces	1.24	–	–	–	1.59	–	1.10	–	1.44	1.36	–	0.03
U	–	–	–	–	–	0.040	0.88	–	–	–	1.83	–	1.67	–	0.40	1.99	–	3.25
W	–	–	–	–	–	1.68	5.17	–	–	–	4.52	–	4.90	–	2.87	6.32	–	2.26
Sn	–	–	–	–	–	2.02	4.79	–	–	–	2.99	–	2.37	–	1.53	3.86	–	3.11
Mo	–	–	–	–	–	0.88	2.23	–	–	–	1.99	–	1.78	–	1.71	3.85	–	2.09
La	9.72	6.76	5.08	11.25	0.69	1.55	23.02	14.68	36.04	27.84	23.56	–	16.40	–	6.29	15.89	25.76	14.36
Ce	17.00	17.88	11.00	25.48	1.70	4.78	44.78	35.55	67.82	55.98	47.5	–	32.36	–	11.61	26.58	49.84	32.19
Pr	2.57	1.97	1.54	3.39	0.31	0.97	5.07	4.67	7.48	6.26	5.44	–	3.59	–	1.35	2.75	5.50	3.90
Nd	10.62	8.09	7.05	14.19	1.61	5.90	18.96	19.34	25.77	22.53	20.82	–	14.22	–	5.01	9.52	19.33	14.99
Sm	2.17	1.90	1.79	3.21	0.56	2.21	3.53	3.89	3.97	3.74	4.18	–	2.70	–	0.10	2.19	3.04	3.27
Eu	0.76	0.53	0.81	0.87	0.26	0.81	0.75	1.24	0.68	0.88	1.11	–	0.75	–	0.29	0.26	1.13	0.38
Gd	2.59	2.31	2.37	3.97	0.81	2.95	2.63	3.71	3.64	3.89	3.59	–	2.27	–	0.77	2.25	2.84	3.64
Tb	0.35	0.36	0.33	0.59	0.13	0.45	0.40	0.52	0.34	0.45	0.51	–	0.31	–	0.11	0.36	0.33	0.43
Dy	1.95	2.15	1.97	3.38	0.78	2.96	2.01	2.96	1.55	2.33	2.8	–	1.70	–	0.48	2.26	1.83	1.82
Ho	0.43	0.49	0.43	0.77	0.17	0.72	0.44	0.60	0.24	0.43	0.54	–	0.36	–	0.10	0.53	0.34	0.28
Er	1.03	1.24	1.06	1.90	0.42	1.62	1.06	1.45	0.56	0.98	1.35	–	0.88	–	0.24	1.31	0.77	0.56
Tm	0.14	0.18	0.15	0.27	0.05	0.24	0.14	0.20	0.07	0.12	0.19	–	0.14	–	0.036	0.23	0.10	0.07
Yb	0.96	1.18	1.00	1.82	0.35	1.53	1.01	1.29	0.47	0.77	1.24	–	1.04	–	0.25	1.57	0.64	0.47
Lu	0.15	0.18	0.16	0.28	0.05	0.22	0.16	0.18	0.07	0.10	0.18	–	0.15	–	0.043	0.25	0.08	0.07

at the Instituto de Geofísica and Instituto de Geología, UNAM, Mexico City, detailed analytical procedures are described here. Between 5 and 25 kg of fresh sample material was crushed in a jawbreaker, ground in a disk mill, aliquotised and finally pulverised with a tungsten carbide mill set. Biotite and muscovite minerals were separated from the 0.2–0.3 mm WR powder fraction and then further separated with a Frantz isodynamic magnetic separator. Garnets from sample LCB-9 were handpicked after magnetic separation. Between 20 and 100 mg of WR sample powder and mineral concentrates were dissolved in Teflon bombs with hydrofluoric acid and HCl. Solutions were split into two aliquots. One was spiked with an ^{87}Rb spike, the other with a combined ^{84}Sr–^{145}Nd–^{149}Sm spike for isotope dilution analysis. Rb, Sr, and rare earth elements (REE) were separated using quartz-glass columns with Dowex cation exchange resin, calibrated by atomic absorbtion spectroscopy. Sm and Nd separations were performed with a different set of smaller columns filled with Teflon powder coated with hydrogen di-ethylhexyl-phosphate. These columns were calibrated colorimetrically. Rb isotope ratios were measured with an NBS-type single collector mass spectrometer (Teledyne Model SS-1290) in peak-jumping mode. Sr, Sm, and Nd isotope ratios were measured with a Finnigan MAT 262 mass spectrometer equipped with a variable multicollector system (eight Faraday cups) in static mode. Samples were loaded as chlorides on double rhenium filaments and were measured as metallic ions. Sr and Nd element concentrations and isotopic ratios were determined with the same runs of six blocks, each of ten individual isotopic ratios. For Sm four blocks of five individual isotopic ratios were measured, and for Rb between 20 and 40 $^{85}Rb/^{87}Rb$ ratios were measured. The results were averaged and are listed together with the standard deviation (1σ) of the runs in Table 2 (for Sr and Nd). Sr-, Nd-, and Sm-isotopic ratios were corrected for mass fractionation by normalising to $^{86}Sr/^{88}Sr = 0.1194$, $^{146}Nd/^{144}Nd = 0.7219$ and $^{152}Sm/^{147}Sm = 1.783\,08$. Total blanks were 0.4 ng for Rb, 2.3 ng for Sr, 0.7 ng for Nd and 0.02 ng for Sm.

Rb–Sr WR-mineral ages and WR isochron ages, their 1σ errors and MSWD (Figs. 5 and 6), were calculated after Wendt in Brooks et al. (1968). Sm–Nd model ages (t_{DM}, Table 2) were calculated with parameters for the depleted upper mantle $^{143}Nd/^{144}Nd = 0.513\,089$ and $^{147}Sm/^{144}Nd = 0.2128$ (mean value for peridotitic xenoliths near San Luis Potosí, central Mexico; Schaaf et al., 1994). Decay constants for ^{87}Rb of $1.42 \times 10^{-11}\,a^{-1}$ and for ^{147}Sm of $6.54 \times 10^{-12}\,a^{-1}$ were used for all age calculations.

Major- and trace-element concentrations were determined by X-ray fluoresence (XRF) and inductively coupled plasma–mass spectroscopy (ICP-MS) at LUGIS-UNAM and partly at the CRPG, Vandoeuvre, France (emission-ICP and ICP-MS).

Drill cores were cut into standard samples and the remanence measured in an AGICO JR5 magnetometer. Demagnetisation was carried out using thermal demagnetisers by Schonstedt and Magnetic Measurements and a proprietary alternating field demagnetiser (maximum field: 200 mT).

5. Major and trace elements

The heterogeneous character of the LCB intrusives is illustrated by the large range of their major- and trace-element distributions (Table 1). SiO_2 contents of the granitoids are between 75.2 and 51.5 wt%, Al_2O_3 between 19.0 and 13.7 wt% and MgO between 5.1 and 0.2%. Data from the El Novillo gabbronorites are not included here but are presented in Riewe (1998) and will be published elsewhere. The mean of the available geochemical data from the northern part of the Puerto Vallarta Batholith (Schaaf, 1990) is also given for comparison (Figs. 2 and 3).

Major elements show typical calc-alkaline differentiation trends, for instance decreasing iron concentrations with increasing SiO_2 contents (Table 1), mainly due to the early crystallisation of magnetite. In Fig. 2 the LCB granitoids plot in the medium- and high-K field (Peccerillo and Taylor, 1976), with a tendency for the deformed SiO_2-rich granitoids to be more K_2O-enriched. Regarding trace-element concentrations, all LCB

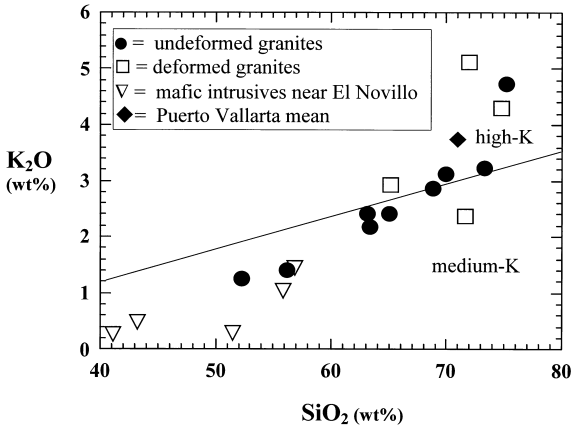


Fig. 2. Plot of K_2O (wt%) versus SiO_2 (wt%) for the LCB intrusives. The boundaries between high-K and medium-K fields are those of Peccerillo and Taylor (1976). Also shown is the mean value for the granitoids of the northern Puerto Vallarta Batholith.

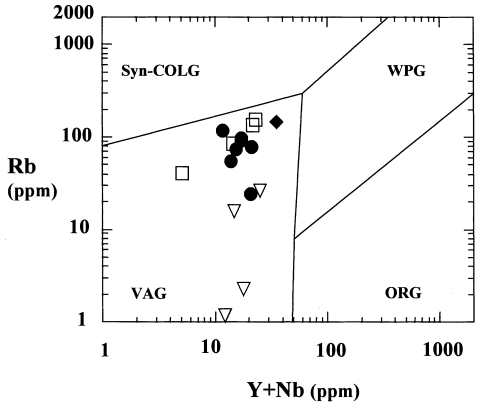


Fig. 3. Rb (ppm) versus Y + Nb (ppm) discrimination diagram for granites (Pearce et al., 1984). LCB intrusives and Puerto Vallarta mean value plot in the volcanic-arc granites (VAG) field. Syn-COLG: syn-collisional granites; WPG: within-plate granites; ORG: ocean-ridge granites. Symbols are the same as in Fig. 2.

samples plot together with the Puerto Vallarta samples in the Volcanic Arc Granite field (Fig. 3).

C1 chondrite-normalised REE distributions are given in Fig. 4. Data from the undeformed granitoids (Fig. 4a) and from the deformed intrusives (Fig. 4b) are fairly similar, displaying moderate to strong enrichment of the LREE (30–150 times chondritic) and weak enrichment of the HREE (2–10 times chondritic). Some samples, especially

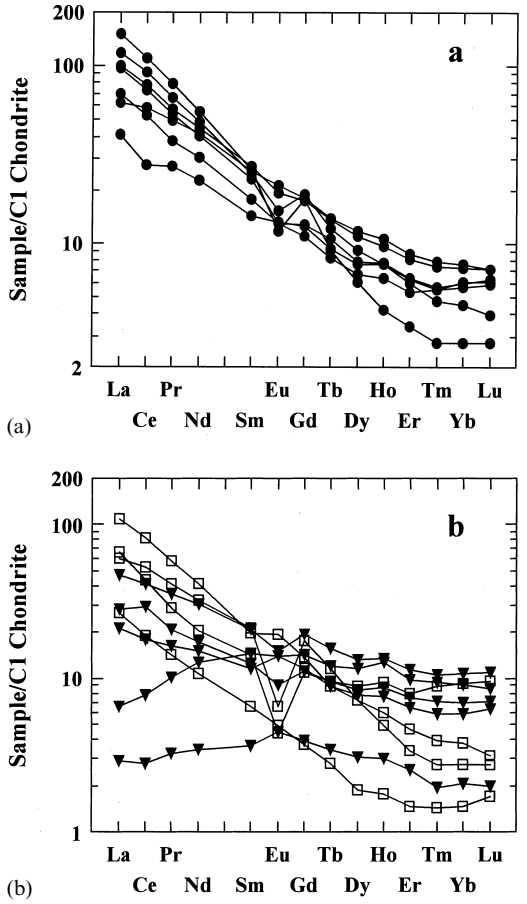


Fig. 4. C1 chondrite-normalised REE distribution diagrams for the LCB intrusives. (a) Values for seven undeformed granitoids. (b) Values for four deformed granitoids (open squares) and for five mafic intrusives (black triangles) near the El Novillo complex.

from the deformed group, have large negative Eu anomalies.

A third group of samples is defined by the mafic, mainly tonalitic, deformed and undeformed granitoids near the El Novillo Complex (Fig. 4b). Tonalitic samples from that area are generally less enriched in LREE with almost equal HREE distribution in comparison with the prior groups. Also included are a gabbronorite (LCB-8A) and a hornblendite (LCB-8B) collected from these tonalites The gabbronoritic sample is a xenolith of the El Novillo Complex; the hornblendite, more probably, is regarded as a cumulate of the mafic, mainly

tonalitic magma, as supported by the enrichment of the LREE, indicating a primary formation of hornblende.

In summary, the LCB intrusives clearly belong to the calc-alkaline, medium- to high-K magmatic suite of a destructive active continental margin with variable amounts of crustal contamination.

6. Isotopic results

Table 2 gives the results of the Rb–Sr and Sm–Nd analyses from 18 WR intrusive samples (both undeformed and deformed), three biotite preparations from undeformed granitoids, and one biotite, white mica, and garnet separation from a diatexitic (deformed) sample.

6.1. Geochronology

Biotite-WR ages were obtained by the Rb–Sr method from undeformed intrusives from the south-central part of the LCB [biotite-granite LCB-19A; 90 ± 2 Ma (1σ errors)], from the northern part southeast of La Paz (tonalite LCB-3; 93 ± 2 Ma), and west of the El Novillo Complex (tonalite LCB-7; 116 ± 2 Ma). Biotite- and muscovite-WR Rb–Sr and (pyrope-almandine)

Table 2
Rb–Sr and Sm–Nd isotopic compositions and element concentrations of the LCB intrusives (WR and corresponding mineral samples are shown in bold face). Isotope values are adjusted to $^{87}Sr/^{86}Sr = 0.710\,23$ for the NBS 987 $SrCO_3$ and to ϵ-Nd = 0 for the BCR-1 standards. Relative reproducibilities (1σ) of the isotope ratios of these standards are ± 0.03‰ and ± 0.022‰. The relative uncertainties in $^{87}Rb/^{86}Sr$ are $\pm 2\%$ (experience-based value for the LUGIS-laboratory) and in $^{147}Sm/^{144}Nd$ $\pm 0.5\%$ (1σ). Relative reproducibilities (1σ) for Rb, Sr, Sm and Nd element concentrations are $\pm 1.5\%$, $\pm 0.7\%$, $\pm 1.6\%$ and $\pm 1.2\%$ respectively. Age values in last column are mineral-WR cooling ages

Sample #	Rock/ mineral type	Rb (ppm)	Sr (ppm)	Sm (ppm)	Nd (ppm)	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr \pm 1\sigma$	$^{87}Sr/^{86}Sr_i$ (90 Ma)	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd \pm 1\sigma$	ϵ-Nd	ϵ-Nd_i (90 Ma)	t_{DM} (Ga)	Age (Ma)
Undeformed granitoids E La Paz														
LCB-3	**Tonalite**	**55.1**	**450.7**	**3.6**	**13.8**	**0.354**	**0.703 949 ± 31**	**0.703 496**	**0.157**	**0.512 916 ± 17**	**5.42**	**5.84**	**0.47**	
LCB-3	**Biotite**	**253.4**	**3.1**			**244.809**	**1.028 368 ± 203**							**93 ± 2**
Mafic to ultramafic intrusives intruding El-Novillo complex														
LCB-5	Qz-Diorite	15.7	364.6	3.6	14.7	0.124	0.703 777 ± 31	0.703 618	0.150	0.512 898 ± 17	5.07	5.57	0.46	
LCB-6	Qz-Diorite	1.2	523.0	2.0	7.7	0.007	0.703 603 ± 30	0.703 594	0.155	0.512 876 ± 34	4.64	5.09	0.56	
LCB-7	**Tonalite**	**26.0**	**400.1**	**3.4**	**14.3**	**0.188**	**0.703 943 ± 31**	**0.703 703**	**0.143**	**0.512 933 ± 21**	**5.75**	**6.34**	**0.34**	
LCB-7	**Biotite**	**208.9**	**5.2**			**117.289**	**0.897 096 ± 63**							**116 ± 2**
LCB-8A	Gabbro-Norite	0.8	271.1	0.56	1.5	0.009	0.703 555 ± 38	0.703 543	0.218	0.512 977 ± 27	6.61	6.33		
LCB-8B	Hornblendite	2.3	232.4	2.25	5.8	0.028	0.703 620 ± 41	0.703 584	0.234	0.513 047 ± 31	7.98	7.52		
Undeformed granitoids south-central LCB														
LCB-13	Bi-Granite	98.5	419.4	3.5	18.6	0.680	0.706 372 ± 47	0.705 502	0.112	0.512 461 ± 16	−3.45	−2.52	0.95	
LCB-18	Tonalite	24.3	703.1	3.4	19.4	0.100	0.705 550 ± 30	0.705 422	0.128	0.512 588 ± 22	−0.98	−0.22	0.90	
LCB-19A	**Bi-Granite**	**119.1**	**184.5**	**4.7**	**29.7**	**1.868**	**0.708 465 ± 31**	**0.706 076**	**0.097**	**0.512 520 ± 19**	**−2.30**	**−1.19**	**0.75**	
LCB-19A	**Biotite**	**596.5**	**4.0**			**451.298**	**1.281 518 ± 40**							**90 ± 2**
LCB-19C	Granodiorite	92.2	577.4	3.5	17.7	0.462	0.706 228 ± 43	0.705 637	0.119	0.512 522 ± 22	−2.26	−1.41	0.92	
LCB-20	Granodiorite	77.7	473.4	3.7	17.5	0.475	0.707 055 ± 37	0.706 444	0.128	0.512 463 ± 15	−3.41	−2.66	1.12	
LCB-21	Bi-Granite	102.7	340.1	4.9	28.2	0.874	0.707 442 ± 38	0.706 324	0.106	0.512 408 ± 13	−4.49	−3.48	0.97	
LCB-23	Granodiorite	73.6	442.3	2.6	12.9	0.482	0.705 688 ± 35	0.705 072	0.121	0.512 561 ± 21	−1.50	−0.67	0.88	
MS 74	Granodiorite	77.0	621.0	0.8	4.4	0.359	0.705 893 ± 30	0.705 434	0.105	0.512 452 ± 17	−3.63	−2.56	0.90	
Deformed granitoids														
LCB-1	Granite	41.2	750.3	1.2	4.7	0.159	0.705 029 ± 30	0.704 826	0.158	0.512 635 ± 24	−0.06	0.35	1.26	
LCB-9	**diatex. Granite**	**153.7**	**120.4**	**1.8**	**7.7**	**3.695**	**0.711 682 ± 34**	**0.706 957**	**0.143**	**0.512 618 ± 36**	**−0.39**	**0.19**	**1.03**	
LCB-9	**Biotite**	**732.6**	**7.1**			**312.181**	**1.141 551 ± 44**							**98 ± 2**
LCB-9	**Muscovite**	**587.0**	**7.9**			**221.531**	**1.029 746 ± 43**							**103 ± 2**
LCB-9	**Garnet**			**19.5**	**15.7**				**0.750**	**0.513 088 ± 20**				**118 ± 12**
LCB-17	Granodiorite	85.5	470.7	3.4	19.4	0.526	0.706 931 ± 42	0.706 258	0.105	0.512 415 ± 21	−4.35	−3.33	0.95	
LCB-19B	Orthogneiss	133.4	90.9	5.9	26.9	4.256	0.722 272 ± 40	0.716 829	0.132	0.512 329 ± 15	−6.03	−5.32	1.43	

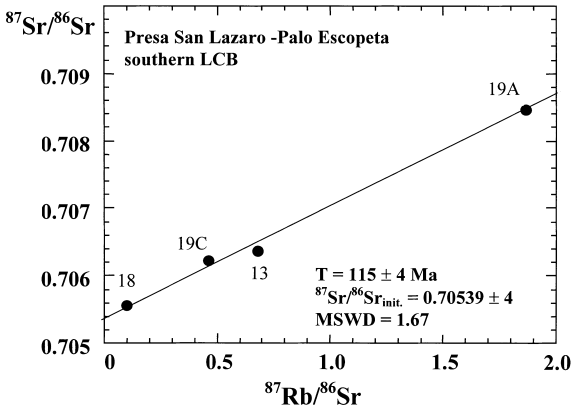


Fig. 5. Rb–Sr isochron diagram for four undeformed granitoid samples from the southern LCB. 1σ errors are smaller than figure labels. Age values and initial $^{87}Sr/^{86}Sr$ ratios, their 1σ errors and MSWD are calculated after Brooks et al. (1968) using the 1σ errors for $^{87}Sr/^{86}Sr$ from Table 2, and for $^{87}Rb/^{86}Sr$ $\pm 2\%$ ($1\sigma_{rel}$).

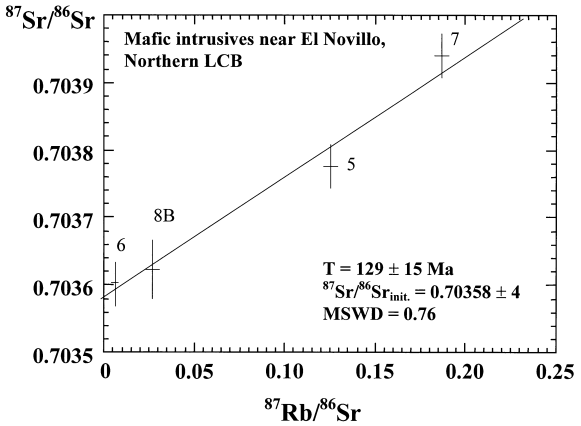


Fig. 6. Rb–Sr isochron diagram for four mafic intrusives near the El Novillo complex. Error bars correspond to the individual $^{87}Sr/^{86}Sr$ 1σ errors and $^{87}Rb/^{86}Sr$ errors of $\pm 2\%$. For explications see Fig. 5.

garnet-WR Sm–Nd ages from a diatexitic granite from the Sierra de la Gata, east-central part of the LCB (sample LCB-9) yielded 98 ± 2 Ma, 103 ± 2 Ma and 118 ± 12 Ma respectively. These ages are cooling-ages, corresponding to closure temperatures of 320 ± 40°C (biotite; Harrison and McDougall, 1980), 500 ± 40°C (muscovite; Jäger, 1979) and 480–700°C (garnet; Humphries and Cliff, 1982). The garnet-WR Sm–Nd age determination in such young material was only possible owing to the high Sm/Nd ratio of 1.25 in the garnets of sample LCB-9. The large (1σ) error in the age value, however, reflects the relatively low growth rate of radiogenic ^{143}Nd in the garnets with respect to the corresponding WR sample in the 118–0 Ma time interval.

An Rb–Sr WR four-point regression line was determined for three undeformed granitic to tonalitic samples from a quarry near Presa San Lázaro (LCB-18, 19A/C, Fig. 1) and one granitic sample 12 km southeast (LCB-13). It is only for these four samples that we have strong evidence for a magmatic consanguinity from within the undeformed LCB suite. The age obtained is 115 ± 4 Ma with an initial $^{87}Sr/^{86}Sr$ ratio of 0.705 39 (Fig. 5).

Four undeformed mafic intrusives surrounding the southern part of the El Novillo Complex are plotted in a second Rb–Sr WR isochron diagram (Fig. 6). Although mafic rocks are generally depleted in Rb, resulting in low Rb/Sr ratios with a small data spread, an acceptable correlation is given for three tonalitic samples and one hornblendite, interpreted by us as a cumulate of the tonalitic sequence. The age is 129 ± 15 Ma with an initial $^{87}Sr/^{86}Sr$ ratio of 0.703 58. Ages and the Sr- and Nd-isotopic data are shown on the map in Fig. 7.

6.2. Sr– and Sm–Nd isotope systematics

Fig. 8 shows an ϵ-Nd versus $^{87}Sr/^{86}Sr$ diagram for the LCB samples, divided into three groups as in Fig. 4. The mafic–ultramafic intrusives near El Novillo plot distinctly within the mantle array field, with extremely primitive ϵ-Nd values between $+5.1$ and $+7.5$ and $^{87}Sr/^{86}Sr$ ratios near 0.7036. Deformed and undeformed granitoids show lower ϵ-Nd values from around 0 to -3.5 and higher $^{87}Sr/^{86}Sr$ ratios between 0.7045 and 0.707, except for the tonalitic sample LCB-3 (E of La Paz) and the migmatic orthogneiss LCB-19B from Presa San Lázaro. The former can be interpreted as a mixture of the mafic to intermediate Sierra Las Cruces granitoids (represented also by sample LCB-1) and the mafic members near El Novillo; the latter gives hints for a more extensive crustal contamination in this sample. Nd model ages (t_{DM}) are around 1 Ga for deformed and unde-

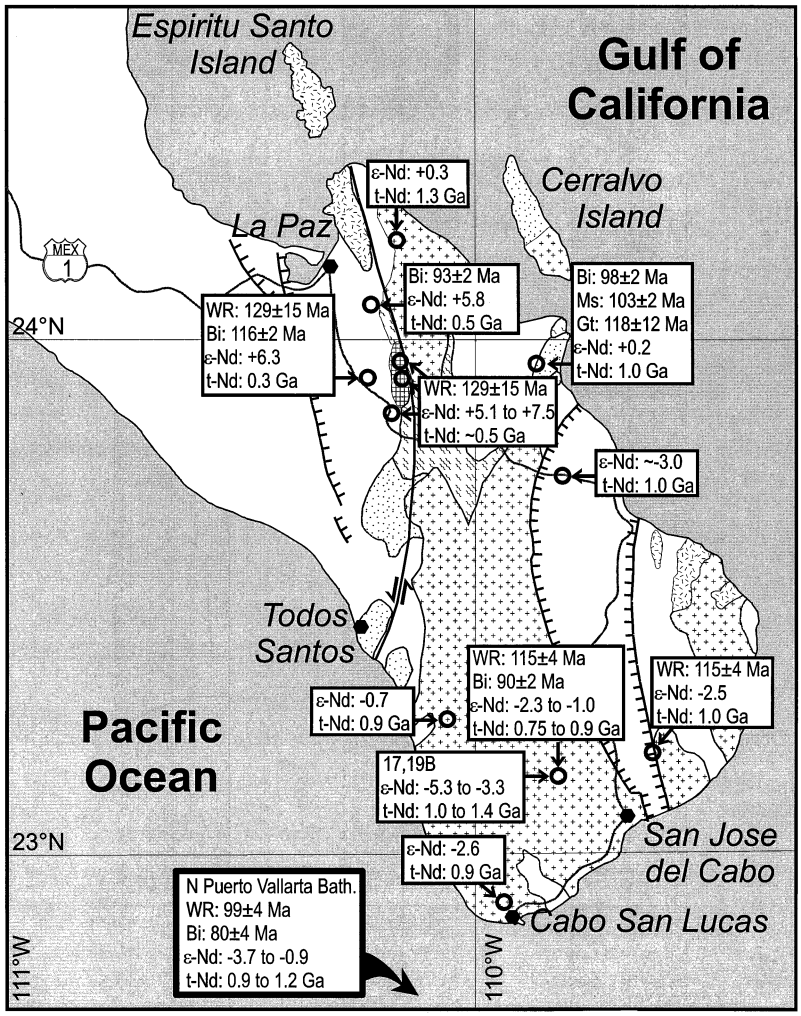


Fig. 7. Distribution of Rb–Sr WR ages (WR), Rb–Sr biotite (Bi)- and muscovite (Ms)-WR ages, Sm–Nd garnet (Gt)-WR age, initial ϵ-Nd values and Nd model ages (t_{DM}) in the LCB. Lithological units and sample numbers are shown in Fig. 1.

formed samples and between 340 and 560 Ma for the mafic tonalite suite. The more evolved silicic rocks have more material of older, still unknown crustal components in their magmas than their primitive counterparts.

7. Palaeomagnetic results

All samples were demagnetised in detail, mainly by the thermal method, as many rocks did not respond well to alternating field demagnetisation. Some samples disintegrated at temperatures above 400–500°C, before complete demagnetisation. Fig. 9 shows typical orthogonal vector demagnetisation plots: samples often exhibited an unstable, probably viscous, overprint, which disappeared at 400–500°C (Fig. 9a); after 400–500°C the demagnetisation curves trend linearly towards the origin. Other samples retained a small high-temperature component that could not be defined owing to erratic directional variations at high temperatures (Fig. 9b). Finally, a number of samples had no stable remanence at all, showing unsystematic vari-

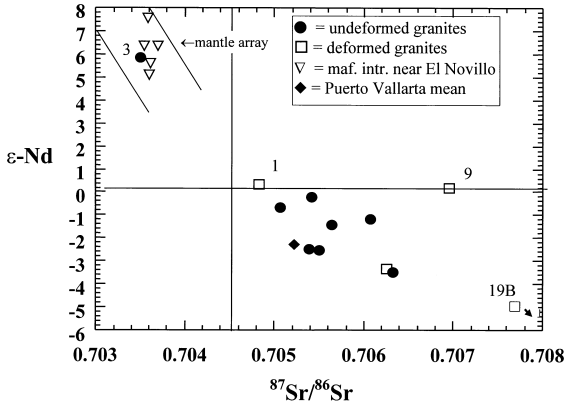


Fig. 8. Initial (calculated for 90 Ma B.P., corresponding to the youngest Rb–Sr biotite-WR cooling age) Nd and Sr isotopic compositions of the analysed LCB intrusives, including the mean value for the northern Puerto Vallarta Batholith (see Table 5). Diatexitic and garnet-bearing granite LCB-9 shows an anomalous high ϵ-Nd value with respect to its $^{87}Sr/^{86}Sr$ ratio, probably due to the introduction of external REE fluids, responsible for garnet growth. Deformed granite LCB-1 and undeformed tonalite LCB-3 are near and within the field of the mafic intrusives near the El-Novillo complex, indicating a possible magmatic consanguinity.

ations in the vector diagrams (Fig. 9c). Therefore, at this moment only results from 18 sites out of 33 are available.

The linear parts of the demagnetisation curves of each sample were chosen as the characteristic remanence directions, and site-mean directions were calculated using Fisher (1953) statistics (Table 3). Three sites (LCB-10, LCB-14, LCB-20) showed large within-site dispersion of directions, with confidence circles $>30^{\circ}$, and are excluded from further interpretation.

Polarity of magnetisation is mixed, with eight reversed and ten normal site-mean VGPs (Table 3). This observation indicates that the remanence was acquired at least partly before or after the Cretaceous Normal Superchron (CNS), which lasted from 118 to 84 Ma (Cande and Kent, 1992). In view of the cooling age determinations mentioned above, the thermomagnetic remanence was probably acquired around the termination of the CNS, possibly between 80 and 90 Ma. The presence of mixed polarities is important, as former studies of PRB rocks had only revealed normal polarity of magnetisation and VGPs close to the

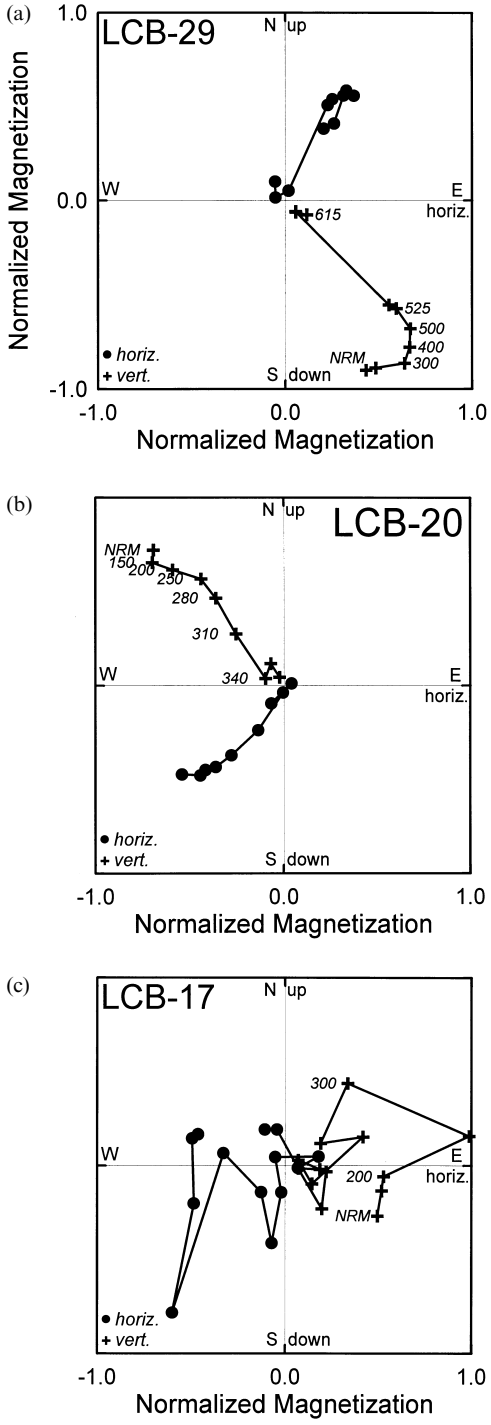


Fig. 9. Normalised orthogonal vector plots for thermal demagnetisation of three representative samples from LCB. Numbers beneath the curves indicate temperatures of demagnetisation steps.

Table 3
Palaeomagnetic data from the LCB. Site codes and coordinates as for isotope sites; th, af, thermal and alternating field demagnetisation; n/r, samples used/rejected for site-mean; R, k, α_{95}, Fisher (1953) parameters; Dec, Inc, declination and inclination of site-mean direction. Sites given in italics are excluded from site mean; sites PB1 to PB4 are from Hagstrum et al. (1985)

Site	Long (°E)	Lat (°N)	Demagnetisation	n	r	R	k	α_{95}	Dec (°E)	Inc (deg)
LCB-1	249.7846	24.1496	th	7	1	6.91730	72.6	7.1	216.1	−44.5
LCB-2	249.7964	24.1963	th	9	0	8.94427	143.6	4.3	212.0	−47.6
LCB-4	249.8480	24.0093	af	5	1	4.66299	11.9	23.2	188.5	−44.7
LCB-6	249.8452	23.9275	af	8	0	7.93886	114.5	5.2	192.8	−34.5
LCB-7	249.7760	23.9277	th	6	1	5.98703	385.4	3.4	24.6	44.7
LCB-10	250.1354	23.9598	th	7	1	3.71734	1.8	64.6	29.8	41.1
LCB-11	249.8999	24.0118	af+th	6	3	5.73774	19.1	15.7	121.5	−58.0
LCB-13	250.3128	23.1471	th	7	1	6.65261	17.3	14.9	182.5	−37.9
LCB-14	250.3128	23.1471	af+th	6	0	3.70392	2.2	60.6	73.4	−17.8
LCB-15	250.3128	23.1471	af	8	0	7.67848	21.8	12.1	35.6	−12.1
LCB-20	250.1765	23.7432	th	12	4	7.03853	2.2	38.8	299.5	67.2
LCB-23	249.9500	23.2586	af+th	7	0	6.64492	16.9	15.1	215.1	−47.6
LCB-24	249.8648	23.2852	th	11	1	10.61821	26.2	9.1	21.6	30.0
LCB-25	249.9963	23.2757	af	6	3	5.83286	29.9	12.4	134.8	−66.5
LCB-26	250.1346	23.6304	th	8	0	7.85964	49.9	7.9	66.1	51.8
LCB-27	249.9242	23.6665	th	9	2	8.83864	49.6	7.4	31.7	38.8
LCB-28	249.9301	23.6676	th	6	0	5.41938	8.6	24.2	18.4	45.3
LCB-29	249.8737	23.6449	th	6	1	5.98703	385.4	3.4	24.6	44.7
PB1	250.1200	23.7300		5		4.95820	96.0	7.9	17.6	45.2
PB3	250.1200	23.7300		4		3.87390	24.0	19.2	343.3	50.9
PB4	250.1200	23.7300		5		4.98620	289.0	4.5	20.3	54.5

North geographic pole [see Hagstrum and Filmer (1990)]. This could be interpreted as the result of a recent magnetic overprint, which was not resolved by the demagnetisation experiments. Our new data probably are not affected by such a process, although we cannot completely exclude an overprint that may have occurred over a time period corresponding to several polarity chrons. Of course, no fold test could be carried out.

Fig. 10 shows the site-mean VGP of our study together with the three VGPs given by Hagstrum et al. (1985). A palaeopole was determined from 17 VGPs, excluding site LCB-11, which falls far from this palaeopole and represents an outlier (McFadden, 1980). For palaeogeographic interpretation the palaeopole is compared with two reference poles for stable North America: the 100 Ma pole of Globerman and Irving (1988) and the 67–97 Ma pole of Gordon and Van der Voo (1995). Using the methods of Beck (1980) and Demarest (1983), we obtain the parameters for northward displacement and rotation as listed in Table 4. With respect to both reference poles, a large clockwise rotation is observed, ranging between $36\pm11^\circ$ and $45\pm9^\circ$. The northward displacement is much smaller, between $6\pm9^\circ$ and $8\pm8^\circ$. Subtracting 2–3° of northward displace-

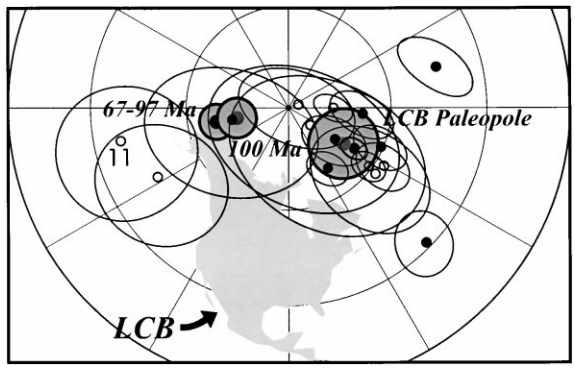


Fig. 10. VGPs for LCB (small dots and circles) with 95% confidence limits (thin ovals) and the corresponding palaeopole (grey shaded confidence limit) in comparison with reference poles for stable North America, for 67–97 Ma and for 100 Ma. Normal and reversed polarity are indicated by dots and circles respectively. Site 11 was excluded as an outlier.

Table 4
Palaeopole for LCB and reference poles for stable North America. Long/Lat, coordinates of pole; A_{95}, confidence limit; R, P, apparent tectonic rotation and poleward displacement after Beck (1980) and Demarest (1983)

Palaeopole	Age	Long. (°E)	Lat. (°N)	A_{95} (deg)	R (deg)	P (deg)
LCB		324.4	71.1	10.1		
North America[a]	100 Ma	192.5	74.1	5.7	39±9	6±8
North America[b]	67–97 Ma	192.0	68.0	5.0	45±9	8±8

[a] Globerman and Irving (1988).
[b] Gordon and Van der Voo (1995).

ment due to the rifting in the Gulf of California, these values become statistically insignificant. The clockwise rotation, on the other hand, remains significant and resembles those values published previously (Hagstrum and Filmer, 1990). A comparison with the early to mid Cretaceous (EMK) reference pole proposed by Dickinson and Butler (1998) leads to very similar results. If we exclude other tectonic movements, the palaeomagnetic data would indicate a large clockwise rotation of the LCB, maybe related to the opening of the Gulf of California, and only a small latitudinal displacement, if any.

8. Discussion

8.1. Geochronologic aspects

The mineral cooling ages and the WR geochronology presented in Section 6 clearly indicate a mid-Cretaceous (Turonian to Hauterivian) intrusion of the LCB granitoids. However, despite field evidence and some geochemical indications, the isotopic data do not yet show unequivocally that the deformed granites are significantly older than the undeformed units, as proposed by Ortega-Gutiérrez (1982) and Sedlock et al. (1993).

The Rb–Sr four-point isochron diagram of cogenetic, undeformed granites from the south-central LCB (Fig. 5) yields an age value of 115±3 Ma ($^{87}Sr/^{86}Sr_{initial}=0.705\,39\pm4$), which we interpret as the time of emplacement of these granitoids. If we add sample MS 74 from Cabo San Lucas to the isochron diagram, the age value remains 115±3 Ma, with an almost equal initial $^{87}Sr/^{86}Sr$ value (0.705 36±4). This is not shown in Fig. 5 as we do not have sufficient knowledge of the geology between Presa San Lázaro and Cabo San Lucas. A more interesting result is obtained when the data of the deformed (diatexitic) granite LCB-9 from the Sierra de la Gata are placed on the isochron diagram: the age value again remains nearly the same with 118±2 Ma ($^{87}Sr/^{86}Sr_{initial}=$ 0.705 36±4). Surprisingly, this age was also obtained for the garnets of sample LCB-9, thus giving evidence for syn-intrusive metamorphic event(s) at least in the eastern part of the LCB (Sierra de la Gata).

Initial $^{87}Sr/^{86}Sr$ and ϵ-Nd values (calculated for 90 Ma B.P.; 0.704 83 to 0.706 96 and −3.5 to +0.3 respectively) and the Nd model ages (0.75 to 1.26 Ga) (Table 2) for the deformed and undeformed intrusives are quite similar, indicative of common source material. An exception is the orthogneissic sample LCB-19B, which seems to be related to a distinct metaigneous event, or which could be part of the minor igneous members of the metasedimentary sequences.

Outstanding features of the quartz-dioritic to tonalitic intrusives in the northern and north-eastern part of the LCB are their very low initial $^{87}Sr/^{86}Sr$ ratios between 0.703 50 and 0.703 62, high initial ϵ-Nd values between +5.1 and +7.5 and low Nd model ages between 0.34 and 0.56 Ga. These rocks are comparable to M-type granites (Pitcher, 1982) and much more primitive than the rest of the LCB intrusives.

Biotite cooling ages are relatively homogeneous in both undeformed and deformed granitoids (90–93 Ma and 98 Ma respectively). This could be due to a reset of the Rb–Sr system of the deformed granitoids during the intrusion of the undeformed plutons. However, we do not see evidence for such

a process. Taking into account the closure temperatures of biotites and muscovites in the Rb–Sr system (see Section 6.1), we can calculate a cooling rate for sample LCB-9 of about 36°C Ma^{-1} for the 500–320°C interval. This rate is comparable to a rate of 45°C Ma^{-1} for the 530–300°C interval in the Puerto Vallarta Batholith (Schaaf, 1990).

Apatite fission track ages for central LCB granitoids carried out by Calmus et al. (1999) are slightly younger, with age values between 68 and 71 Ma. They could indicate a prolonged period of high heat flow within the LCB or later uplift episodes.

8.2. Palaeogeographic implications

Concerning the palaeomagnetic data, no large-scale northward displacement of the LCB is required since the mid-Cretaceous, although it cannot be excluded in view of the error limits (Table 4). Further work is in progress, especially to identify possible (differential?) tilting effects of the LCB pluton. Currently, no geological criteria unequivocally indicate large amounts of post-intrusive tilt. Aranda-Gómez and Pérez-Venzor (1986) pointed out that Miocene volcanic sequences covering the crystalline basement of Espíritu Santo island (N of La Paz, Fig. 1) dip approximately 10° northwest. Quaternary conglomerates forming alluvial fans along the eastern flank of the LCB are also inclined 10–15° to the northwest. It is unknown if these dips are due to regional or local tilting processes or of primary origin. East-up tilting and asymmetric uplift of crystalline basement as reported by Fletcher et al. (1998) may also indicate a complex tilting and uplift history of the LCB. Correction of an eastward-up tilting would displace the LCB palaeopole such as to increase the clockwise rotation. Therefore, and because of the uncertain nature of the observed dip of the overlying Tertiary rocks, no tilt correction was applied to the LCB. Independent data are needed to determine if the LCB was tilted after remanence acquisition and how such a tilt would affect the position of the palaeopole.

The clockwise rotation of about 40° calculated from our data for the LCB exceeds significantly the value of 10–15°, which may be explained by the rifting in the Gulf of California, if Baja has been displaced as a rigid terrane. The value is similar to, although slightly larger than, most of the data reported previously (Hagstrum and Filmer, 1990; Lewis and Stock, 1998). These rotations are clearly different from data obtained from intrusive rocks in southern Mexico (Böhnel et al., 1992), where no clockwise and only a few counter-clockwise rotations were found. Therefore, the tectonic environment of Baja and southern Mexico appears to have been different. That difference may be restricted to the style of local vertical axis rotations; apparently no latitude displacements have occurred in either region. Differences in local crustal block rotations could be the response to predominantly left-lateral shear along southern Mexico on the one hand, and to the opening of the Gulf of California and associated right lateral shear on the other hand.

As a modern analogue to the rifted Gulf of California we may consider the Afar region in east Africa, although the triple junction there consists of three rifts whereas in Mexico it is defined by a rift and subduction zone. The Gulf of Aden spreading centre has been propagating into the continent since about 2 Ma, causing rifting and extension within the continent. This has led to about 11° clockwise rotation of microplates since that time (Acton and Stein, 1991). Such or similar processes, therefore, may have contributed to the apparent rotation of the LCB.

Our geochemical and isotopic data from the LCB intrusives are in good agreement with the data obtained for the granitoids of the Puerto Vallarta Batholith. This large igneous body forms part of the Jalisco Block and has a mid-Cretaceous intrusion age (91–99 Ma) that is only slightly younger than the LCB intrusion events.

Major-element, trace-element, and isotopic data for Puerto Vallarta rocks are included as mean values in Figs. 2 and 8 and plot well within the LCB sample fields. The isotopic data for both intrusives are also compared in Table 5. A good coincidence can be observed for intrusion and cooling ages, cooling rates, initial Sr and Nd isotopic ratios and Nd model ages, especially when

Table 5
Comparison of the isotopic data from the LCB and Puerto Vallarta Batholith. The best coincidence is observed between the southern LCB and the northern Puerto Vallarta granitoids

	Southern LCB, undeformed	Northern LCB, undeformed El Novillo surroundings	Northern Puerto Vallarta Batholith	South-central Puerto Vallarta Batholith	Western Puerto Vallarta Batholith
Intrusion age (Ma)	115 ± 4	129 ± 15	99 ± 4[a]	91 ± 4[a]	n.d.
Rb–Sr biot. WR age (Ma)	90 ± 2	116 ± 2	$78–82 \pm 2$[a]	$75–84 \pm 2$[a]	$75–83 \pm 2$[a]
U–Pb zircon age (Ma)	94[b]	n.d.	101 ± 3.4; 103 ± 6.5[a]	n.d.	n.d.
$^{87}Sr/^{86}Sr_{initial}$	0.705 39	0.703 58	0.705 22[c]	0.704 42[c]	ca 0.704 00[c]
ϵ-$Nd_{initial}$	-3.5 to -0.2	$+5.1$ to $+7.5$	-3.7 to -0.9[a]	0.0 to $+2.0$[a]	-1.6 to $+3.2$[c]
t-Nd_{DM} (Ga)	0.9–1.1	0.3–0.6	0.9–1.2[a]	0.6–0.7[a]	0.5–0.8
Cooling rate (ΔT)	36°C Ma^{-1} (500–320°C)[d]	n.d.	45°C Ma^{-1} (530–300°C)[c]	n.d.	n.d.
Δ Intr. age/Bi age (Ma)	25	13	ca 19	ca 11.5	n.d.

n.d.: not determined.
[a] Data from Schaaf et al. (1995).
[b] Data from Frizzell et al. (1984).
[c] Data from Schaaf (1990), temperature interval defined by the closure temperatures of hornblende and biotite in the K–Ar system (Harrison and McDougall, 1980).
[d] Temperature interval defined by the closure temperatures of muscovite and biotite in the Rb–Sr system (see Section 6.1).

comparing the data from the southern LCB and those from the northern Puerto Vallarta Batholith.

The Puerto Vallarta Batholith displays some more mafic, generally tonalitic outcrops in its western part and some undeformed to slightly deformed high silica, two-mica granites in the eastern part, intruding a greenschist-facies volcanosedimentary sequence of probable late-Triassic to early Jurassic age.

The Puerto Vallarta granitoids are generally less tectonised, except some syn-intrusive 'granite-tectonics', related to the emplacement of the granitic melts and some post-intrusive mafic dikes.

Southeast of Puerto Vallarta, continental margin granites show decreasing intrusion ages down to 22 Ma near the Isthmus of Tehuantepec, lower SiO_2 contents, lower $^{87}Sr/^{86}Sr$ isotopic ratios and higher ϵ-Nd values, combined with lower Nd model ages (Schaaf, 1990; Schaaf et al., 1995). The only granitic body that is petrographically comparable to the LCB intrusives is the partly pseudo-rapakivi textured Acapulco granite. However, this granite has a middle Eocene (44 Ma) intrusion age (Guerrero-García, 1976; Schaaf et al., 1995) and can be ruled out as a comagmatic counterpart of the LCB.

9. Conclusions

By comparing the geochemical and isotopic data of the LCB intrusive rocks with those from the Puerto Vallarta Batholith a magmatic consanguinity is strongly suggested. Together with the palaeomagnetic results of our study, an independent palaeogeographical evolution of the LCB with respect to the rest of Baja California peninsula may have occurred, at least considering the palaeomagnetic data from the peninsula so far published. The La Paz Fault could be the possible accretional structure of the LCB, although alternative sutures may be hidden below Tertiary rocks further to the north.

Gastil (1983) characterised the continental magmatic arc of northwestern and western Mexico, including the LCB, as 'Pluton Composition Belts', starting with gabbros next to the trench and followed by tonalitic, granodioritic and granitic coast-parallel belts. This model is generally in accordance with our observations. Further work to confirm this model is in progress, as is the work necessary to clarify in more detail the relation between the deformed and undeformed intrusives of the LCB. This investigation must include intensive fieldwork

and it is desirable to obtain some zircon U–Pb ages from the deformed intrusives.

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